

CHE 396 Quiz 2

Huda Usman

TOTAL POINTS

92 / 100

QUESTION 1

11 17 / 20

- ✓ - **0 pts** Correct
- **20 pts** wrong or not done
- **0 pts** Click here to replace this description.
- **3** Point adjustment
- 💬 splitting is confusing. no cooler in the below?

QUESTION 2

22 20 / 20

- ✓ - **0 pts** Correct
- **17 pts** only LK and HK identified.
- **20 pts** wrong or not done
- **5 pts** HK calculation is wrong.

QUESTION 3

33 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** reflux ratio not mentioned.
- **2 pts** reflux ratio stated as decreased.

QUESTION 4

44 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** Incorrect

QUESTION 5

55 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** Incorrect

QUESTION 6

66 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** Incorrect

QUESTION 7

77 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** incorrect

QUESTION 8

88 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** incorrect

QUESTION 9

99 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** Incorrect

QUESTION 10

1010 10 / 10

- ✓ - **0 pts** Correct
- **10 pts** Incorrect

QUESTION 11

1111 10 / 15

- **0 pts** Correct
- **15 pts** Wrong or not done
- **10 pts** Not explained well
- ✓ - **5 pts** Pressure not mentioned
- **5 pts** C4s are not mentioned

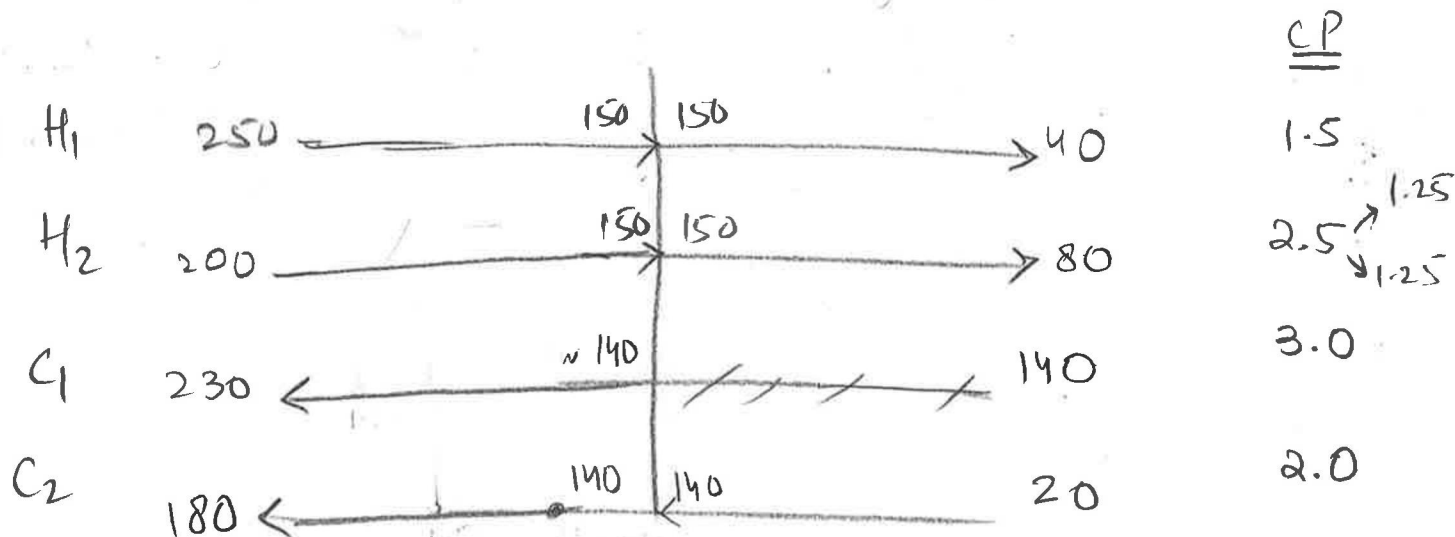
Huda Usman

Name

1. Determine the minimum utility requirements for the process set out below. Take the minimum approach temperature as 10°C . Design a heat exchanger network to achieve maximum energy recovery. Pinch point for the problem is 145°C .

$$\Delta T_{\min} 10^{\circ}\text{C}, T_{\text{pinch}} = 145^{\circ}\text{C}$$

Stream No.	Actual temperature ($^{\circ}\text{C}$)		Heat capacity flow rate CP ($\text{kW}/^{\circ}\text{C}$)
	Source	Target	
1	250	40	1.5
2	140	230	3
3	200	80	2.5
4	20	180	2

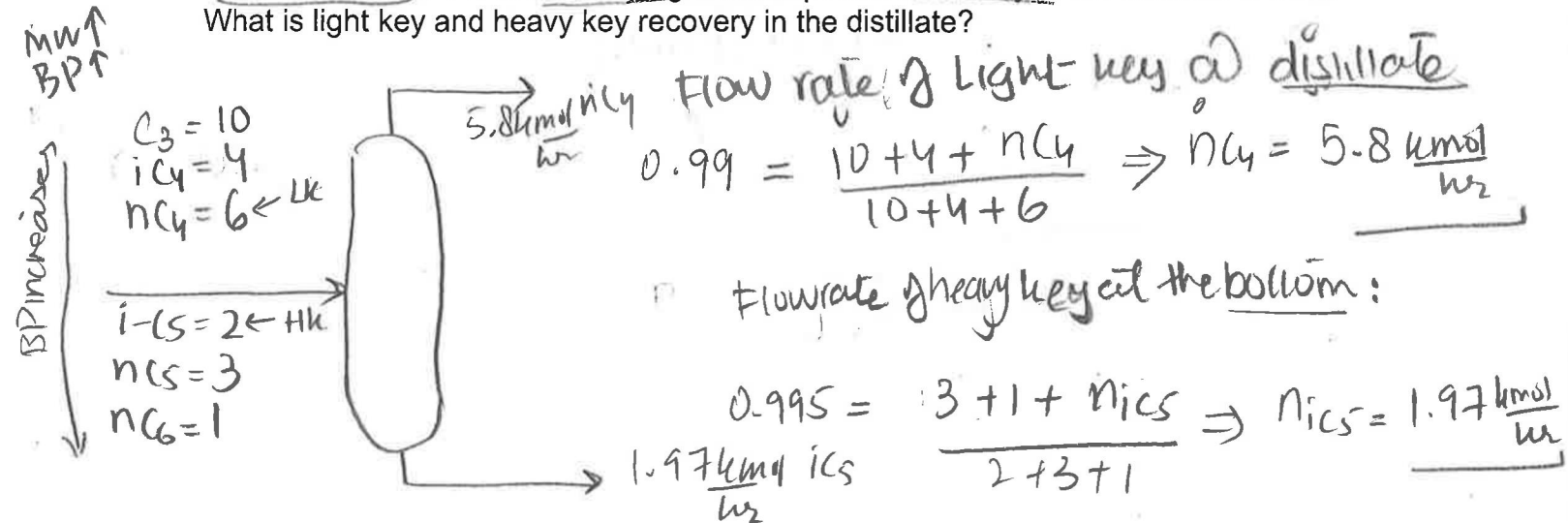


$$T_p(\text{hot}) = T_{\text{pinch}} + \frac{\Delta T_{\min}}{2} = 145 + 5 = 150^{\circ}\text{C}$$

$$T_p(\text{cold}) = 145 - 5 = 140^{\circ}\text{C}$$

Continued on the back of the quiz

2. The composition of the feed to a debutanizer is as follow; propane (C_3) is 10 kmol/h, isobutane ($i-C_4$) is 4 kmol/h, n-butane ($n-C_4$) is 6 kmol/h, isopentane ($i-C_5$) is 2 kmol/h, normal pentane ($n-C_5$) is 3 kmol/h and normal hexane ($n-C_6$) is 1 kmol/h. C_5 recovery in the bottoms is 99.5% and C_4 and lighter compounds recovery in the distillate is 99%. What is light key and heavy key recovery in the distillate?



Recovery of light key in distillate

$$R = \frac{5.8}{6} \times 100 = 96.67\%$$

Recovery of heavy key @ bottom;

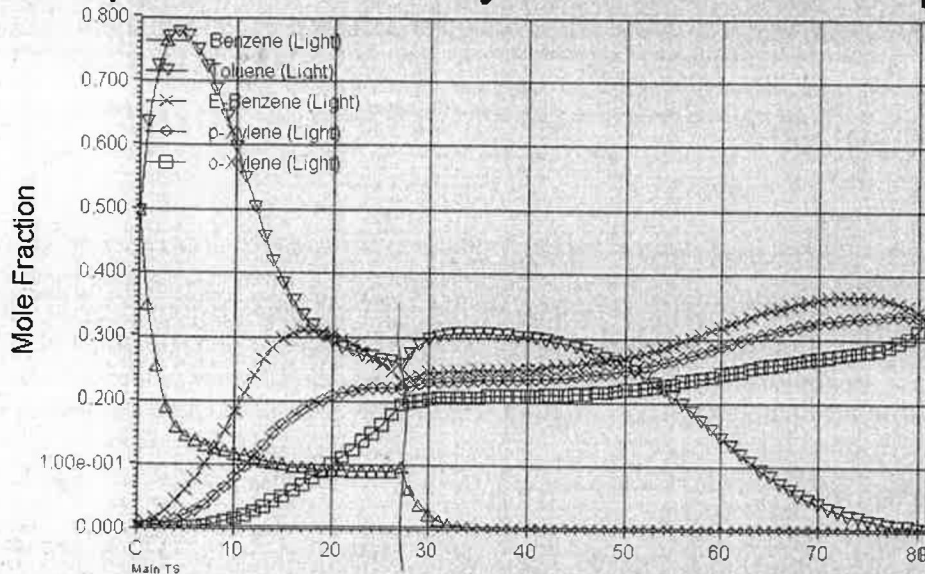
$$R = \frac{(2 - 1.97)}{2} \times 100 = 1.5\%$$

∴ Light key $\Rightarrow$ highest B.P @ the top therefore L.K is $n-C_4$

∴ Heavy key $\Rightarrow$ lowest B.P @ the bottom, therefore H.K. is $i-C_5$ in our case -

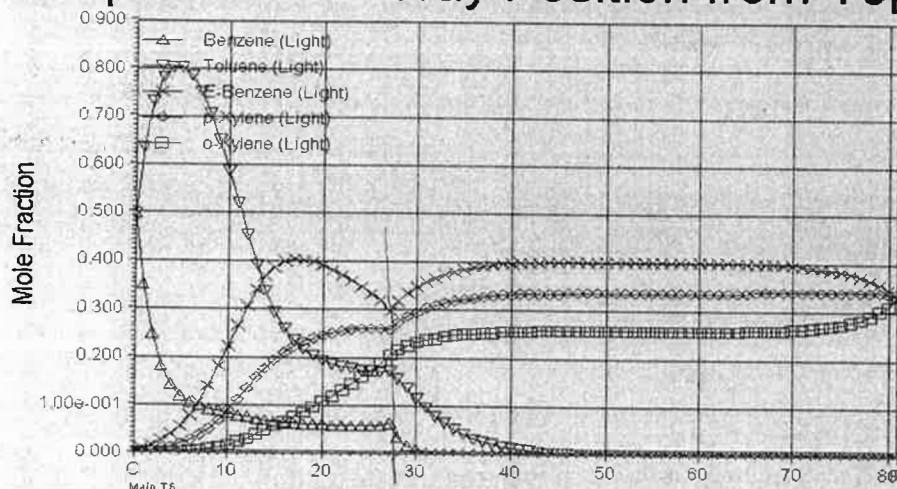
3. Equimolar mixture of benzene, toluene, ethylbenzene (EB), orthoxylene (OX) and paraxylene (PX) is separated by distillation. Composition vs tray position was graphed as:

Composition vs. Tray Position from Top



Later some parameters in the rigorous column has been changed and graph below is generated. Based on the graph below, which parameter has been changed, decreased or increased?

Composition vs. Tray Position from Top



I think above, reflux ratio is increased and is too high therefore the product purity has also increased. The # of stages and the feed location is kept constant. As the reflux ratio increases, the separation also increases which in turn increases the recovery of product @ the bottom.

4. Distillation processes can only separate two species ("A" and "B") if:

- A. They have similar molecular weights.
- B. They have very different enthalpies of vaporization
- ☒ C. They have different vapor pressures.
- D. One species exists as a vapor (at STP), while the other species must be a liquid.

5. The pressure in a tray column

- A. Is higher at the top.
- ☒ B. Is higher at the bottom.
- C. Depends on the type of tray and the tray design.
- D. b and c.
- E. None of the above.

6. The temperature in a tray column

- A. Is higher at the top.
- ☒ B. Is higher at the bottom.
- C. Is the same on every tray.
- D. Could be a or b, depending on the components involved.
- E. None of the above.

7. If a column design results in one that is too tall, which of the following is a possible remedy?

- A. Decrease the reflux ratio.
- ☒ B. Reduce the number of trays.
- C. Decrease the tray spacing.
- D. b and c.
- E. All of the above.

8. Which of the following is true about the feed to a distillation column?

- A. The column feed determines the column pressure.
- ☒ B. The column feed must be larger than the column pressure.
- C. The column feed tray is always in the middle of the column.
- D. b and c.
- E. None of the above.

9. Which of the following is true about the feed location in a distillation column?

- A. Feed on the middle tray.
- B. The feed should be at the tray with approximately the same temperature.
- C. The feed should be at the tray with approximately the same composition.
- ☒ D. b and c.
- E. All of the above.

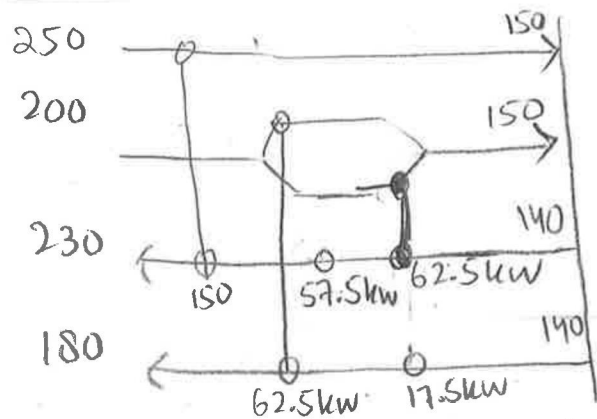
10. What was the pressure of MTBE column in your mini-project 2 design?

We used 27 bar to run the distillation column.

11. Methanol boiling point is higher than MTBE boiling point at 1 atm. And they form azeotrope. However, in your mini-project problem 2, MTBE is separated from methanol by distillation and MTBE is the bottom product. Explain how that column works.

Methanol forms an azeotrope with butane ^{and that} ~~that~~ resultant mixture have lower point than MTBE itself therefore we were getting the methanol as light key at the distillate and MTBE as heavy key at the bottom.

above pinch $C_{ph} \leq C_{pc}$



$$\Delta H_1 = 1.5(150 - 250) = -150 \text{ kW}$$

$$\Delta H_{2a} = 1.25(150 - 200) = -62.5 \text{ kW}$$

$$\Delta H_{2b} = 1.25(150 - 200) = -62.5 \text{ kW}$$

$$\Delta H_3 = 3.0(230 - 140) = 270 \text{ kW}$$

$$\Delta H_4 = 2.0(180 - 140) = 80 \text{ kW}$$

3 heat exchangers, 2 heaters.

below pinch $C_{ph} \geq C_{pc}$

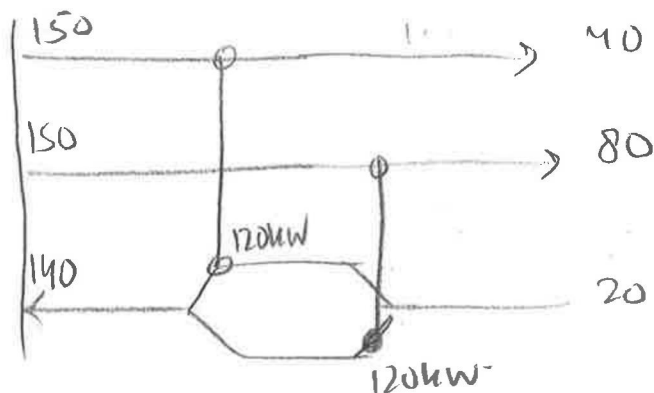
$$\Delta H_1 = 1.5(-150 + 40) = +165 \text{ kW}$$

$$\Delta H_2 = 2.5(-150 + 80) = -175 \text{ kW}$$

$$\Delta H_{3a} = (140 - 20) \times 1 = +120 \text{ kW}$$

$$\Delta H_{3b} = 1 \times (140 - 20) = +120 \text{ kW}$$

2 heat exchangers.



Hence,

5 heat exchangers and 2 heaters are required for the process.

CP
1.5
2.5
3.0
2.0
continued
problem

CP

1.5

2.5

2.0

1.0 1.0